Social Media Engagement Analytics

1.PROJECT OVERVIEW:

This project analyses social media engagement data using Python, focusing on user demographics, post attributes, and interaction metrics. It involves data cleaning, exploration, statistical analysis, and feature engineering to uncover engagement patterns. Visualizations with Matplotlib, Seaborn, and Plot present insights into post-performance, audience behaviour, and content strategy. The goal is to transform raw data into actionable insights for optimizing social media engagement.

2. Project Objectives :

1.Data Quality Improvement

- Clean null values, duplicates, and inconsistent formatting.
- Standardize categorical variables .

2.Engagement Insights

- Identify which **post types, categories, and sentiments** drive higher engagement.
- Compare engagement across **countries and demographics** (age, gender).
- Highlight patterns in likes, comments, shares, and impressions.

3.Statistical Understanding

- Summarize **central tendencies** (mean, median, mode) and **variability** (variance, standard deviation).
- Detect **skewness and distribution patterns** in engagement metrics.
- Use percentiles to understand audience behavior ranges.

4.Visualization & Reporting

- Build clear, interactive dashboards using **Matplotlib, Seaborn, and Plotly**.
- Highlight trends such as **daily engagement**, **post category performance**, and **device-based engagement**.

- Provide visual storytelling for stakeholders.

5.Business Application

- Deliver **actionable recommendations** for improving social media strategy.
- Help stakeholders understand **audience behavior** and optimize content for maximum reach.
- Guide decisions on which content types and strategies yield the best ROI.

3.DATA INGESTION:

Data source: *social media engagement.csv*

Data from: *GitHub*

Time line:*2022-2023*

4.TOOLS USED:

Python (NumPy, pandas)

Visualization: seaborn, matplotlib

5.Data Inspection

1. *Loaded the dataset into a Pandas DataFrame and viewed the first and last five records using head() and tail().*
2. *Checked the dataset dimensions, column names, data types, and structure using shape, columns, dtypes, and info().*
3. *Identified missing values and duplicate records using isnull().sum() and duplicated().sum().*
4. *Generated summary statistics and analyzed numerical relationships using describe() and the correlation matrix.*
5. *Examined unique values, counted category frequencies, and summarized data using unique(), nunique(), value_counts(), and groupby().*

6.Data Cleaning

1. *Checked for missing values and handled them appropriately.*
2. *Removed duplicate records to improve data quality.*
3. *Converted columns to the correct data types.*

4. *Verified and standardized the dataset for consistency.*
5. *Prepared the cleaned dataset for analysis and visualization.*

7.FEATURE ENGINEERING

Added this column for better module accuracy

- *A new field called engagement_score was created by summing the number of likes, comments, and shares for each post, providing a consolidated measure of overall engagement.*
- *Another derived field, log likes, was generated using the natural logarithm transformation (np.log1p) on the likes column to normalize the distribution and reduce skewness in the data.*

8.Data Transformation

1. *Converted data types of numerical and date columns.*
2. *Formatted the posted_at column to datetime format.*
3. *Standardized categorical values for consistency.*
4. *Created derived columns for better analysis (if applicable).*
5. *Prepared the transformed dataset for visualization and analysis.*

9. EXPLORATORY DATA ANALYSIS:

- ***Find correlations between columns*** *to understand relationships among numeric variables.*
- ***Detect outliers and anomalies visually*** *using plots such as boxplots, scatterplots, or histograms.*
- ***Compare groups and categories*** *with group by summaries and categorical visualizations.*
- ***Identify time-based trends in data*** *by analysing engagement metrics across dates and timelines.*

10.Statistical Analysis:

- *Calculated summary statistics.*
- *Measured mean, minimum, and maximum values.*

- *Analyzed data distribution.*
- *Examined correlations between variables.*
- *Compared grouped data.*
- *Identified key trends and patterns.*

11. DATA VISUALIZATION:

MATPLOTLIP:

- *Scatter Plot: Likes vs. Impressions*
- *Line Plot: Daily Engagement Trend*
- *Bar Chart: Posts by Category*
- *Pie Chart: Gender Distribution*
- *Histogram: Age Distribution*
- *Box Plot: Engagement Rate Distribution*

SEABORN:

- *Count Plot: Post Type Count*
- *Bar Plot: Average Likes by Category*
- *Violin Plot: Followers vs. Sentiment*
- *Pair Plot: Numeric Features (likes, comments, shares, impressions, engagement rate)*
- *Heatmap: Correlation Matrix*
- *Strip/Swarm Plot: Engagement Rate vs. Device Type*

PLOTLY (INTERACTIVE):

- *Line Chart: Engagement Trend (with range slider)*
- *Bar Chart: Average Likes by Category (sorted)*
- *Scatter/Bubble Chart: Likes vs. Impressions (Combined Subplots Dashboard: Engagement Trend, Avg Likes by Category, Likes vs. Impressions Bubble Chart*

12.CONCLUSION:

The analysis shows that Video posts deliver the strongest engagement, while countries like India and France lead in impressions. Negative sentiment posts slightly outperform others, sparking more interaction.

Verified accounts maintain steadier performance, and mobile devices drive higher watch times. Overall, optimizing content type, sentiment strategy, timing, and regional focus will maximize engagement and support data-driven decision making.